Xingyang Li

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EDUCATION

Massachusetts Institute of Technology (MIT) — Cambridge, MA
Visiting Student at MIT HAN Lab, EECS Department — 07/2025 – 12/2025
Advisor: Prof. Song Han

Shanghai Jiao Tong University (SJTU) — Shanghai, China
B.Eng. in Computer Science & Technology — 09/2022 – 06/2026 (Expected)
Member of (ACM Honors Class), an elite CS program for top 5% students

- **GPA** (All): **4.03**/4.3, Ranking: **3**/30, 19 A+ courses
- Selected core courses: Computer Vision: 100/100, Machine Learning: 97/100, Comprehensive Design for Computer System: 97/100, Information Theory: 98/100, Advanced Compiler Project: 100/100, Linear Algebra: 100/100

RESEARCH INTERESTS

I study efficient machine learning, with a focus on sparse neural networks and low-bit model quantization. My long-term vision is to help build efficient superintelligence—models that stay compact, accessible, and reliably scalable to serve increasingly broad and demanding real-world needs.

SELECTED PUBLICATIONS

Radial Attention: $\mathcal{O}(n \log n)$ Sparse Attention with Energy Decay for Long Video Generation
Xingyang Li, Muyang Li*, Tianle Cai, Haocheng Xi, Shuo Yang, Yujun Lin, Lvmin Zhang, Songlin Yang, Jinbo Hu, Kelly Peng, Maneesh Agrawala, Ion Stoica, Kurt Keutzer, Song Han*
Annual Conference on Neural Information Processing Systems (NeurIPS), 2025. [arxiv] [code] [website] [blog]

Voyager: Real-Time Splatting City-Scale 3D Gaussians on Resource-Constrained Mobile Devices
Zheng Liu, He Zhu*, Xingyang Li, Yirun Wang, Yiming Gan, Wei Li, Yujiao Shi, Jingwen Leng, Minyi Guo, Yu Feng*
Under review at ICLR 2026. [website]

SLTarch: Towards Scalable Point-Based Neural Rendering by Taming Workload Imbalance and Memory Irregularity
Xingyang Li, Jie Jiang*, Yu Feng, Yiming Gan, Jieru Zhao, Zihan Liu, Jingwen Leng, Minyi Guo*
International Conference on Computer-Aided Design (ICCAD), 2025; (**Oral Presentation**). [arxiv]

Harnessing Conventional Video Processing Insights for Emerging 3D Video Generation Models: A Comprehensive Attention-aware Way
Tianlang Zhao, Jun Liu*, Xingyang Li*, Li Ding, Jinhao Li, Shuaiheng Li, Jinbo Hu, Guohao Dai*
Design Automation Conference (DAC), 2025; (**Oral Presentation**). [pdf]

RESEARCH EXPERIENCE

Massachusetts Institute of Technology, MIT HAN Lab — Cambridge, MA
Advisor: Prof. Song Han

Quantization Algorithm for Efficient Video Diffusion Models — 08/2025 – 11/2025

- We leverage activation similarity to improve the quantization of video diffusion models. (*first author, submitted to CVPR 2025*)

Efficient and Scalable Video Diffusion Models with Sub-quadratic Complexity — 02/2025 – 07/2025

- We presented the first work of leveraging sparse attention for longer video generation, with a sub-quadratic static sparsity mask and minimal LoRA fine-tuning. (*first author, accepted by NeurIPS 2025*)

- I observed the "Spatiotemporal Decay" phenomenon in the video diffusion transformer, which states that attention score and density decays as spatial and temporal token distance grow, shedding light on sparsity mask design for attention-based models.
- I designed the $\mathcal{O}(n \log n)$ attention mask for efficient video generation without any tuning, and further extended video length with LoRA fine-tuning.
- Our Radial Attention mask achieved up to 1.9× speedup for default-length video generation, and offered up to 4.4× training cost saving and inference acceleration for up to 4× longer videos.
- Our open-source repository achieves **550+ GitHub Stars**, and has been integrated into the state-of-the-art video diffusion model inference engine.

Shanghai Jiao Tong University, DAI Lab Shanghai, China
Advisor: Prof. Guohao Dai

Mixed Precision Video Diffusion Transformers with Hardware Accelerator 10/2024 – 12/2024

- We presented an algorithm-hardware co-design for efficient video generation models. (*co-first author, accepted by DAC 2025*)
- I designed an attention importance speculation algorithm with a mixed-precision computation strategy for the self-attention operator, which is the bottleneck of video diffusion models.
- We achieved 1.45× speedup and 1.63× energy efficiency compared to existing accelerators.

Shanghai Jiao Tong University, EPCC Lab Shanghai, China
Advisor: Prof. Yu Feng and Prof. Jingwen Leng

Scalable and Low-Latency Mobile Rendering System for 3D Gaussian Splatting 01/2025 – 05/2025

- We presented an effective solution to enable city-scale 3DGS rendering on mobile devices. (*third author, under review at ICLR 2026*)
- I designed a temporal-aware algorithm to reuse the level-of-detail tree traversal result with kernel prototypes, alleviating the latency bottleneck on the tree traversal stage.
- We achieved over 100× less data transfer and up to 8.9× speedup while retaining rendering quality.

Algorithm-Hardware Co-design for Scalable 3D Gaussian Splatting 08/2024 – 04/2025

- We presented an algorithm-architecture framework for scalable 3DGS, which is bottlenecked by level-of-detail tree traversal and rasterization. (*first author, accepted by ICCAD 2025*)
- I designed a subtree-based tree traversal algorithm with BFS-based re-ordering and the specialized hardware architecture, ensuring streaming DRAM accesses.
- We achieved 3.9× speedup over mobile GPUs and 1.8× speedup over existing accelerators.

HONORS

2025 Commercial Sponsorship Scholarship (**Top 6** at Zhiyuan College, SJTU)
2025 SenseTime Scholarship (**30 winners nationwide** with aspiration in AI research)
2024 Commercial Sponsorship Scholarship (**14 winners** each year in SJTU)
2023 Longfor Merit Scholarship (**Top 10** at Zhiyuan College, SJTU)
2022, 2023, 2024 Zhiyuan Honorary Scholarship (**Top 2%** in SJTU)
2023, 2024 Academic Excellence Scholarship (Ranked 4-th, 2-nd respectively in ACM Class of 2026)

TECHNICAL SKILLS

Language: TOFEL: Total 115 (Reading 30, Listening 30, Speaking 25, Writing 30); CET 6: 665/710
Programming and Software: Python, C++, CUDA, Bash, Git, Java, Go, and Verilog